NOTES-WISE



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Table of Contents

List of Figures **vi**

Certificate 1

LETTER OF UNDERTAKING 3

EXORDIUM 4

ACKNOWLEDGEMENTS 5

Abstract 6

Introduction **7**

0.1 Background 8

0.2 Problem Statement 8

0.3 Aim and Objectives 9

Literature Review **11**

0.4 Literature Review 11

Proposed Solution **14**

0.5 Functional and Non-Functional Requirements 14

0.5.1 Functional Requirements 14

0.5.2 Non-Functional Requirements 15

0.6 Development Methodology 16

0.7 Design Diagrams 17

0.7.1 Use Case Diagram 17

0.7.2 Class Diagram 19

0.7.3 Entity Relationship Diagram 20

0.7.4 Sequence Diagram 21

0.7.5 System Design Diagram 22

0.8 Project Timeline 23

0.8.1 Gantt Chart 23

Software Components **24**

0.9 Development Framework and Tool 24

0.9.1 Tool 24

0.9.2 Framework 24

0.10 Python and Js Libraries 24

0.10.1 Js Libraries 24

0.10.2 Python Libraries 25

0.11 AI Models 26

0.11.1 Grammar Model 26

0.11.2 Summarization Models 26

0.11.3 QuizCraft Models 26

0.11.4 Chatbot Model 27

Results and Discussion **28**

0.12 Testing and Results 28

0.13 Discussion and Recommendations 35

0.13.1 Discussion 35

0.13.2 Recommendations 35

Use-Case Catalogue **37**

0.14 Authentication 37

0.15 Workspace & File Management 38

0.16 Text Notes 41

0.17 Diagram Notes 42

0.18 Flashcards 43

0.19 Smart Note Tools 44

0.20 AI Chatbot 45

0.21 Cloud Storage & Access 45

Future Tasks **47**

0.22 Audio Notes 47

0.22.1 Record and Save Audio 47

0.22.2 Convert Audio to Text 47

0.23 Collaborative Notes 48

0.23.1 Real-Time Collaboration 48

0.24 Note Tagging & Search Filters 48

0.24.1 Tag Notes 48

0.24.2 Advanced Search 48

0.25 Daily Planner Integration 49

0.25.1 Create Study Schedule 49

0.26 Offline Editing Mode 49

0.26.1 Local Editing with Sync 49

Conclusion and Bibliography **50**

0.27 Conclusion 50

Bibliography 51

List of Figures

1 Use Case Diagram 18

2 Class Diagram 19

3 Entity Relationship Diagram 20

4 Sequence Diagram 21

5 System Design Diagram 22

6 Gantt-chart 23

7 Authentication Page 28

8 Home Page 29

9 CreateNotes Page 29

10 CheckNotes Page 30

11 Document Page 30

12 Diagram Page 31

13 FlashCards Page 31

14 QuizCraft Page 32

15 ChatBot Page 32

16 Summarizer Page 33

17 Grammar Result 33

18 ContactUs Page 34

19 Admin Page 34

Certificate

It is certified that the contents and the form of the report submitted by **Rana Umair Babar** (Registration Number: **F22-BSc-IET-049**), **Shuaib Waheed Kayani** (Registration Number: **F22-BSc-IET-047**), and **Muhammad Hamza** (Registration Number: **F22-BSc-IET-031**) have been found satisfactory for the degree of Bachelor of Information Engineering Technology (BSc. IET) requirements.

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LETTER OF UNDERTAKING

We, the undersigned, hereby confirms that the Project we have provided is solely our own effort. WE did not copy our report partially or completely from any other student or from any other source either against payment or free and we did not provide any plagiarized material in any section of our report. WE further confirm that the documents (Job Confirmation Letter, etc.) that we have provided are genuine (i.e. not forge/fake) and have been issued by the authorized person in the organization. If we are found guilty of misstating, misleading or concealing the facts about our activities (either academic or non-academic but relevant to this course) at any stage, the university is authorized to take disciplinary action against us according to university policies and regulations.

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EXORDIUM



Praise be to Allah, Lord of Creation, The Compassionate, the Merciful, King of Judgment-day!

You alone we worship, and to You alone we pray

for help,

Guide us to the straight path

The path of those who You have favored,

Not of those who have incurred Your wrath, Nor of those who have gone astray.

ACKNOWLEDGEMENTS

We would like to express our sincere gratitude to everyone who contributed to the completion of this project and report. A special thanks to our project supervisor, Ms. Huma Israr, who guided us throughout the project, shared her invaluable knowledge, and suggested different ideas that helped in our webapp project.

A special thanks to the members: Umair, Shuaib, and Hamza, who collaborated on every stage of the project. Our hard work and dedication were instrumental in ensuring that we successfully developed and implemented this webapp. We worked together in harmony, each contributing our strengths and ideas to bring this project to life.

Abstract

Notes-Wise is a web application designed to transform the note-taking experience by addressing the challenges of traditional methods. In a fast-paced digital world, where students and professionals alike are increasingly relying on online platforms, Notes-Wise offers a seamless solution for organizing and managing notes while improving writing skills. The app integrates features such as real-time grammar and spelling correction, task management, and resources for mastering the English language, making it an all-in-one tool for learners. Notes-Wise aims to create a productive learning environment, empowering users to take clearer, more organized notes and develop their English writing proficiency, from basic grammar rules to advanced techniques. Unlike other note-taking platforms, Notes-Wise incorporates educational tools that facilitate the learning process and provide ongoing feedback to users, turning note-taking into an opportunity for continuous growth. The project also acknowledges the growing reliance on digital platforms like Microsoft Word, Grammarly, and Evernote, which have set the stage for improving learning and writing outcomes. Ultimately, Notes-Wise seeks to provide a comprehensive solution for note-taking and language learning, enhancing productivity and the overall educational experience for students, professionals, and anyone striving to enhance their English skills.

Introduction

Note-taking is a helpful tool that improves clarity of thoughts and aids in memorization, allowing people to process [1] information more easily and remember important ideas for future use. In this context, technology has changed to make the note-taking experience better. Notes-Wise is a new web app that tackles the problems of traditional note-taking by offering a simple solution for creating and organizing notes effectively. It addresses issues like clutter and poor feedback, helping users improve their writing skills and create a more productive learning environment. With its easy-to-use interface, you can quickly jot down ideas, memorize your notes, manage tasks, and keep track of important information. One of its key features is automatic mistake correction, which ensures your notes are clear and professional. Beyond note-taking, Notes-Wise serves as a complete learning tool for mastering the English language. From basic grammar rules to advanced writing techniques, the app provides resources for all skill levels [2]. Whether you're taking daily notes, writing in English, or preparing for presentations, Notes-Wise gives you the tools you need to improve your writing skills. With a focus on clarity and organization, this app not only simplifies your note-taking process but also helps you develop important writing skills. Notes-Wise aims to boost productivity and improve the learning experience, making it the perfect companion for students, professionals, and anyone looking to enhance their English

writing skills.

0.1 Background

As society has increasingly shifted away from hardcopy notes, the advantages of digital platforms like Microsoft Word and Grammarly have become clear, offering flexibility, ease of editing, and accessibility that are essential for modern learners [3]. This transition has underscored the importance of self-learning in today's fast-paced world, as online learning provides a wealth of resources that promote independent study and skill enhancement. To address the growing need for effective digital solutions, this app not only corrects grammatical and spelling errors in real-time but also serves as a comprehensive platform for developing writing proficiency. By fostering a learning environment, it helps users identify and understand their mistakes, turning every note into an opportunity for growth. With intuitive editing tools for straightforward revisions and educational resources such as grammar guides and practice exercises, the app supports users in mastering various writing styles. Ultimately, this project aims to empower individuals to take effective notes and enhance their English capabilities, making it an invaluable tool for students, professionals, and anyone committed to improving their writing skills in the digital age [4].

0.2 Problem Statement

Effective note-taking is vital for academic success and professional growth. Traditional methods often lack organization and real-time feedback, impeding clarity and productivity. Existing digital solutions like Grammarly, Evernote, and Microsoft OneNote

focus on specific aspects such as grammar correction or note organization but fail to provide an integrated platform that combines all these features.

Students and professionals struggle to maintain clarity in their notes, manage tasks efficiently, and improve their English writing skills [5]. The available tools do not offer a comprehensive solution that integrates note-taking, language learning, and task management in one seamless interface. Additionally, they lack features like real-time grammar correction and structured educational resources, which are critical for continuous improvement.

There is a need for an all-in-one tool that helps users take better notes, improve their writing proficiency, and manage tasks effectively. Notes-Wise addresses this gap by providing a web application that combines note organization, grammar correction, and learning resources for mastering English. This platform aims to enhance productivity, clarity, and learning by offering an integrated solution for note-taking and language development, benefiting students, professionals, and anyone looking to improve their writing skills.

0.3 Aim and Objectives

Aim: The primary aim [6] of this project is to enhance the note-taking experience through the development of Notes-Wise, a web application designed to improve clarity, organization, and productivity in writing and learning. The app aims to provide a seamless integration of note-taking, language learning, and task management to help users enhance their academic and professional performance.

Objectives:

A. To provide a user-friendly interface that enables users to quickly jot down ideas,

manage tasks, and organize important information efficiently.

B. To facilitate the mastery of the English language by offering resources that cater to various skill levels, from basic grammar rules to advanced writing techniques.

C. To cater to a broad audience, including students, professionals, and anyone looking to enhance their English writing skills, by providing a versatile tool that meets their specific needs.

D. To integrate real-time grammar and spelling correction to help users identify and fix writing mistakes as they take notes.

E. To enable users to manage their tasks and deadlines alongside their notes, improving overall productivity and organization [7].

F. To provide personalized feedback and suggestions for improvement based on the user's writing and note-taking habits.

G. To offer a cloud-based storage solution for users to securely store and access their notes from any device at any time.

H. To allow users to create and organize notes into categories and tags, making it easier to find and review information.

Literature Review

0.4 Literature Review

Effective note-taking and learning tools have evolved significantly with the advent of technology. Traditional methods of note-taking often lack flexibility and can lead to disorganization [8], reducing the effectiveness of the learning process. Digital note-taking solutions have emerged to address these shortcomings, with some platforms focusing on note organization, while others aim to improve writing skills and productivity.

1). Digital Note-Taking Tools

Platforms like **Evernote**, **Notion**, and **Microsoft OneNote** offer solutions for organizing notes across devices, creating a seamless digital experience. Evernote and OneNote are renowned for their cloud-based storage, allowing users to access their notes anywhere, but they lack robust grammar correction features. Notion, while versatile in note organization and project management, does not provide specialized resources for language learning or writing enhancement. These tools, while effective for basic note-taking and organization, fail to offer integrated features for mastering English writing skills.

2). Grammar and Writing Assistance

Grammarly, a widely-used tool for grammar and spell checking, provides real-time feedback on writing, helping users improve sentence structure, punctuation, and grammar. However, while Grammarly excels at language correction, it lacks note-taking functionality, forcing users to rely on separate platforms for note management. Similarly, other language enhancement tools, such as **Hemingway Editor** and **ProWritingAid**, offer features aimed at improving writing clarity, style, and grammar, but do not provide an integrated solution for note-taking.

3). Language Learning Platforms

Apps like **Duolingo** and **Babbel** focus on language acquisition, offering interactive lessons and exercises to improve grammar and vocabulary. These platforms cater to users looking to learn a language from scratch, but they do not support the note-taking aspect that is essential for effective learning. Furthermore, they are not tailored to help users with academic writing or note organization.

4). Integrated Solutions for Learning

Recent research suggests that combining note-taking and writing assistance within a single platform can improve learning outcomes. **TILOVATOVA (2024)** highlighted that digital platforms that integrate language learning resources with note-taking features can significantly enhance students' ability to retain information and improve writing skills. Similarly, **Hoque et al. (2018)** emphasized the importance of offering feedback during the note-taking process, suggesting that real-time error correction can improve both writing clarity and cognitive retention.

Studies in *self-regulated learning* emphasize the importance of continuous feedback and task management in achieving academic success. **Masouleh (2012)** found that apps which provide structured learning materials and feedback, like grammar rules

and practice exercises, have a significant positive impact on learners' ability to self-correct and improve their writing skills.

5). Existing Gaps and the Need for Notes-Wise

Despite the variety of existing tools for note-taking, language learning, and writing assistance, no single platform offers a comprehensive solution that combines all three features. Tools like Grammarly and Evernote excel in their respective domains, but fail to bridge the gap between effective note-taking, task management, and language learning. **Notes-Wise** aims to fill this gap by offering a web application that combines note organization, task management, grammar correction, and educational resources for mastering English writing. It is designed to cater to students, professionals, and anyone looking to improve their note-taking and writing skills.

In conclusion, while many tools address specific aspects of note-taking or writing assistance, the need for an integrated solution that enhances both productivity and writing skills remains unmet. **Notes-Wise** seeks to address these gaps by offering a user-friendly, comprehensive platform that not only helps users organize and take better notes but also provides the resources needed to improve their English proficiency.

Proposed Solution

0.5 Functional and Non-Functional Requirements

The Notes-Wise web application [9] requires both functional and non-functional specifications to meet user needs. Functional requirements include providing a user-friendly interface for note-taking, task management, grammar correction, and English language resources. These features enable users to organize their notes and improve writing skills. Non-functional requirements focus on system performance, such as response time, reliability, security, and usability. The application must be responsive across devices and browsers, ensuring easy access to notes and data security. Notes-Wise aims to deliver a high-performance, secure tool that supports users in enhancing their note-taking and English writing capabilities.

0.5.1 Functional Requirements

A. The system must allow users to create, edit, and delete notes.

B. The application should provide real-time grammar and spelling correction for the notes written by the users.

C. Users should be able to organize their notes in folders or categories for easy retrieval.

D. The app must provide task management features like setting reminders and to-do lists.

E. The system should provide a set of resources for improving English writing, such as grammar guides, vocabulary building exercises, and writing tips.

F. The app should allow users to search for specific notes or tasks.

G. The platform must support multi-user accounts, allowing users to store their personal notes securely.

0.5.2 Non-Functional Requirements

A. The application should be responsive, ensuring it works smoothly on both mobile and desktop devices.

B. It must be compatible with modern browsers (Chrome, Firefox, Safari, Edge).

C. The system should have minimal response time for performing actions like saving notes or loading resources.

D. The application must ensure high security by encrypting user data and preventing unauthorized access.

E. The app should have a user-friendly interface, with easy navigation and clear instructions for new users.

F. The platform should have 99.9

G. It should be scalable, able to handle a large number of concurrent users without degrading performance.

H. The application must be maintainable and allow for future updates or feature additions.

0.6 Development Methodology

The development of the Notes-Wise web application follows the **Agile methodology**, a flexible and iterative approach that emphasizes collaboration, adaptability, and customer feedback [10]. Agile focuses on delivering incremental improvements through short development cycles known as sprints, which allows for rapid adjustments based on user needs and feedback.

Phases of Development:

A. **Requirement Gathering:** In this phase, we identify and document both functional and non-functional requirements of the system, based on input from potential users, stakeholders, and project supervisors.

B. **Design:** The application architecture and user interface are designed to ensure a simple, intuitive, and responsive experience. The focus is on clarity, organization, and seamless integration of the various features such as note-taking, task management, and grammar correction.

C. **Development:** The core development phase follows Agile sprints, where features like the note editor, task manager, and real-time grammar correction are developed incrementally. Regular testing is conducted to ensure quality and functionality.

D. **Testing:** Continuous testing and validation of the application take place at each sprint. Both functional testing (checking individual features) and non-functional

testing (ensuring performance, security, and usability) are performed.

E. **Deployment and Maintenance:** Once the web app is ready, it is deployed for users to access. Maintenance involves ongoing support and updates to enhance functionality and address user feedback.

F. **Documentation:**The Notes-Wise web application will include user and system documentation. The user manual will guide users on note creation, organization, and grammar correction. System documentation will cover the technical aspects, such as architecture, technologies, and APIs. An installation guide will provide steps for deployment and troubleshooting. This documentation ensures an efficient experience for both users and developers.

0.7 Design Diagrams

The design diagram for the Notes-Wise web application visually represents the system's architecture and how different components interact. It includes key elements such as the frontend, backend, and database, highlighting their connections and data flow. The backend handles business logic, while the database stores user data, notes, and learning resources securely. This design ensures a seamless user experience, efficient data handling, and scalability, making the app an effective tool for enhancing English writing skills.

0.7.1 Use Case Diagram

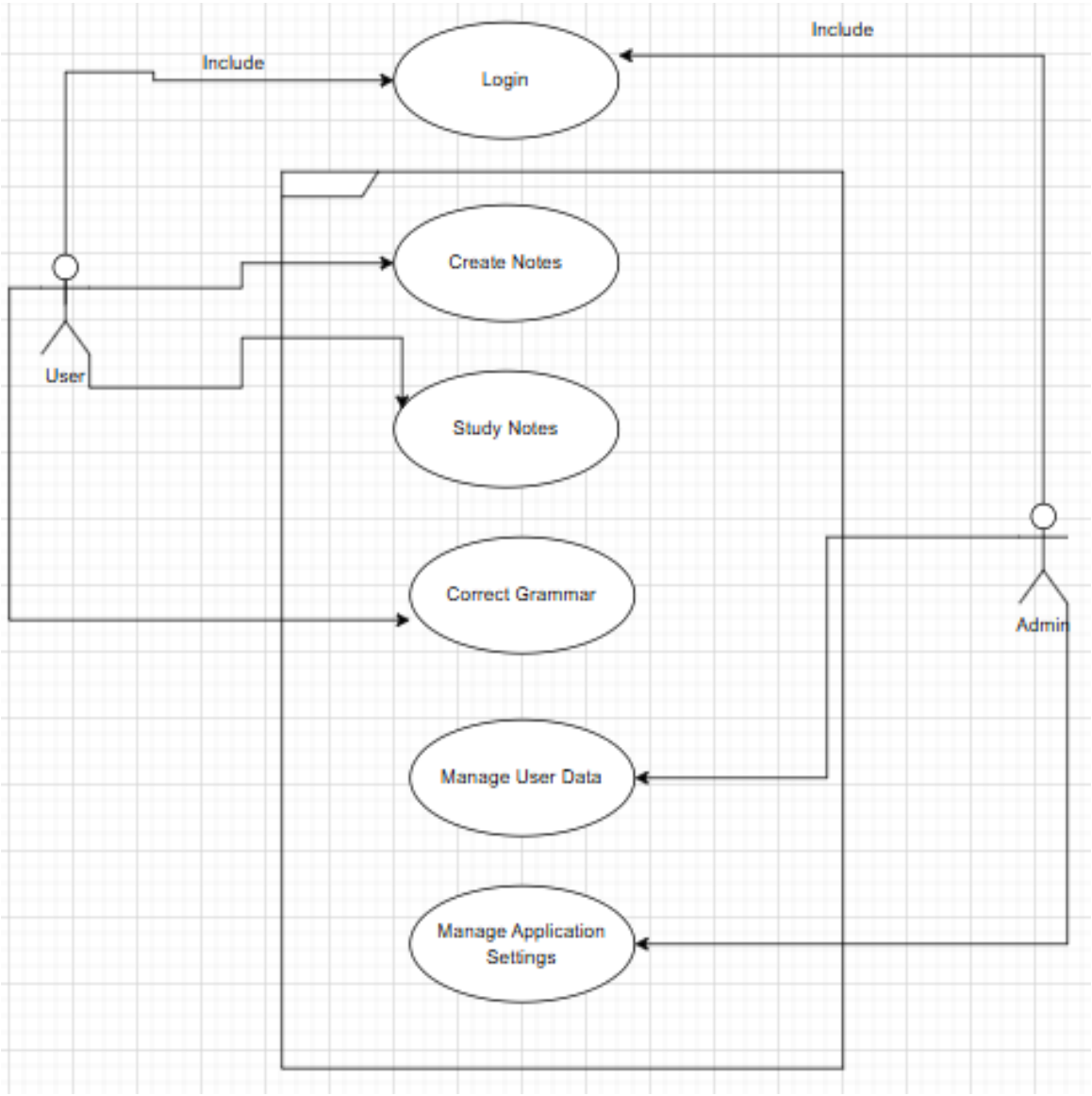


Figure 1: Use Case Diagram

0.7.2 Class Diagram

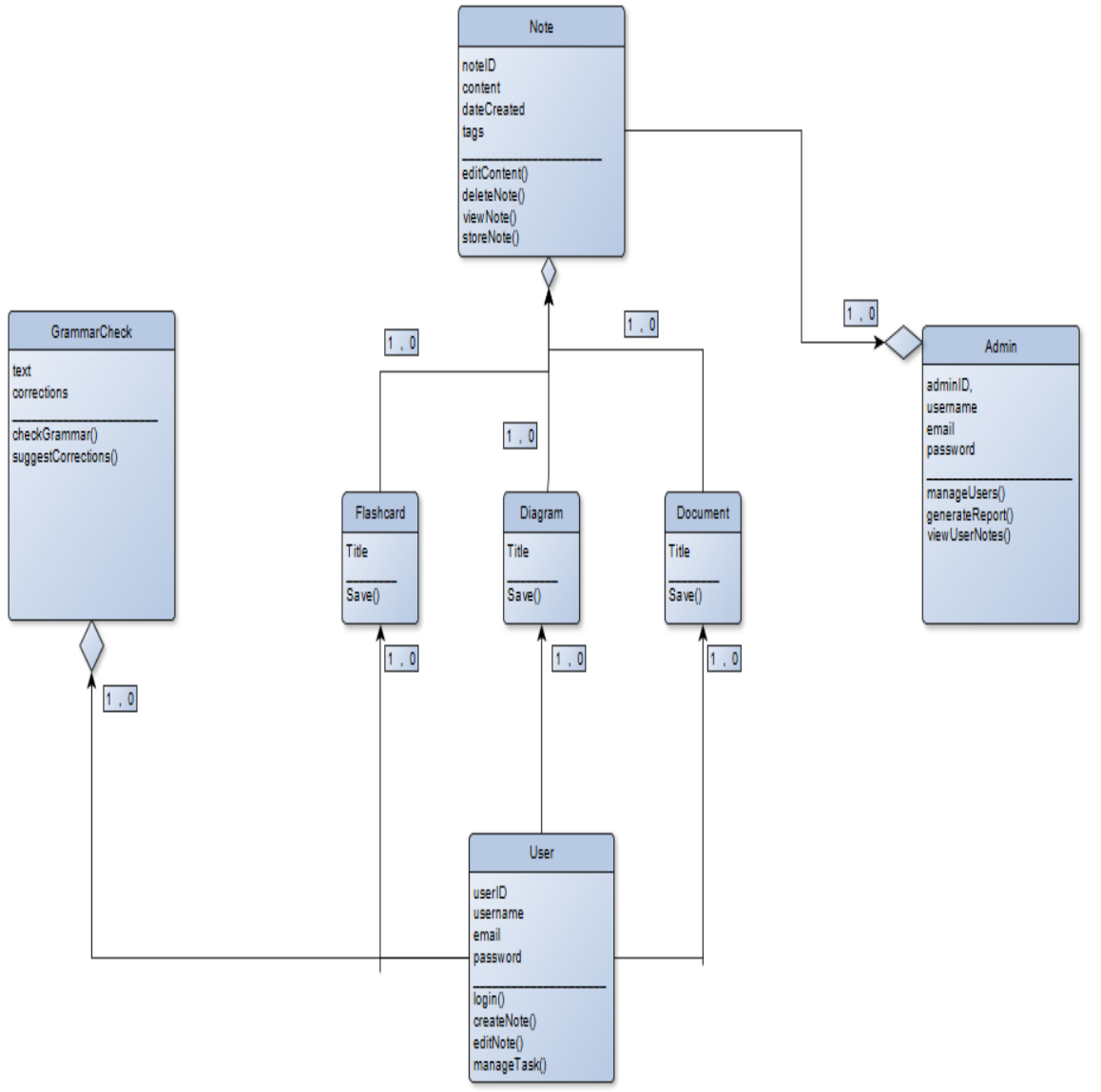


Figure 2: Class Diagram

0.7.3 Entity Relationship Diagram

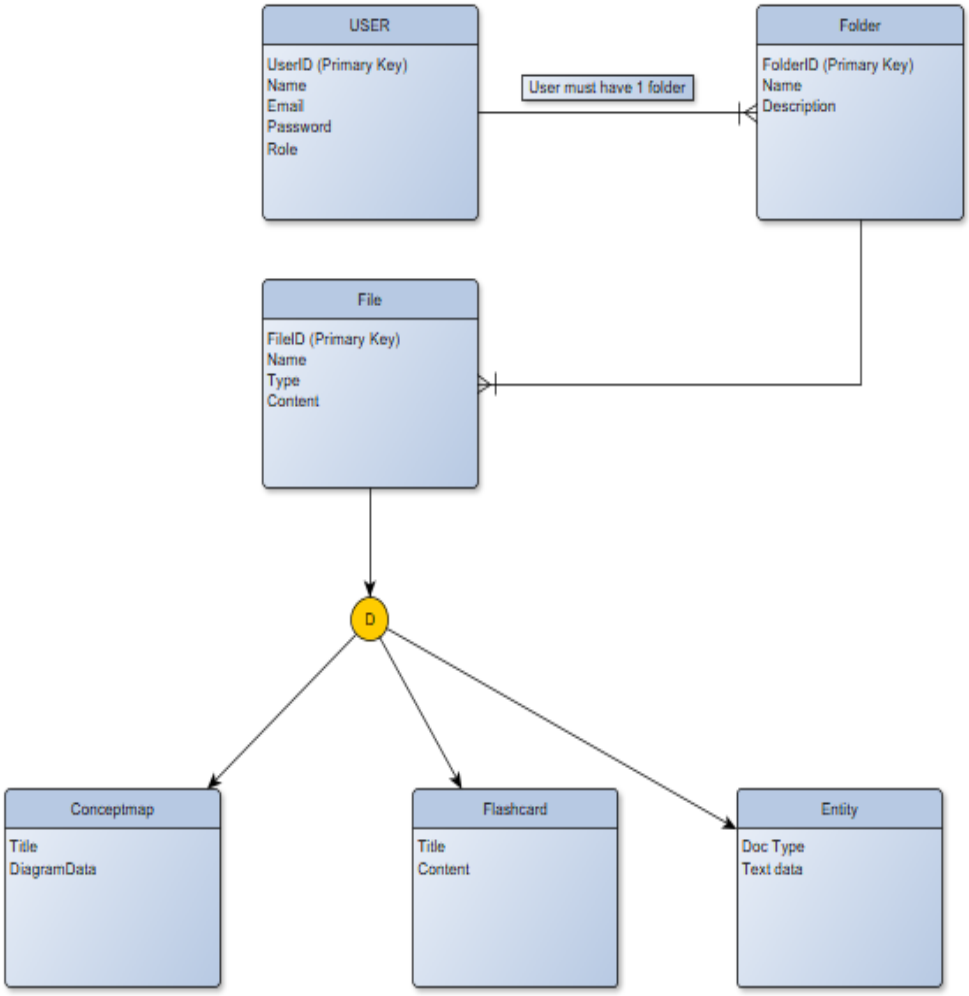


Figure 3: Entity Relationship Diagram

0.7.4 Sequence Diagram

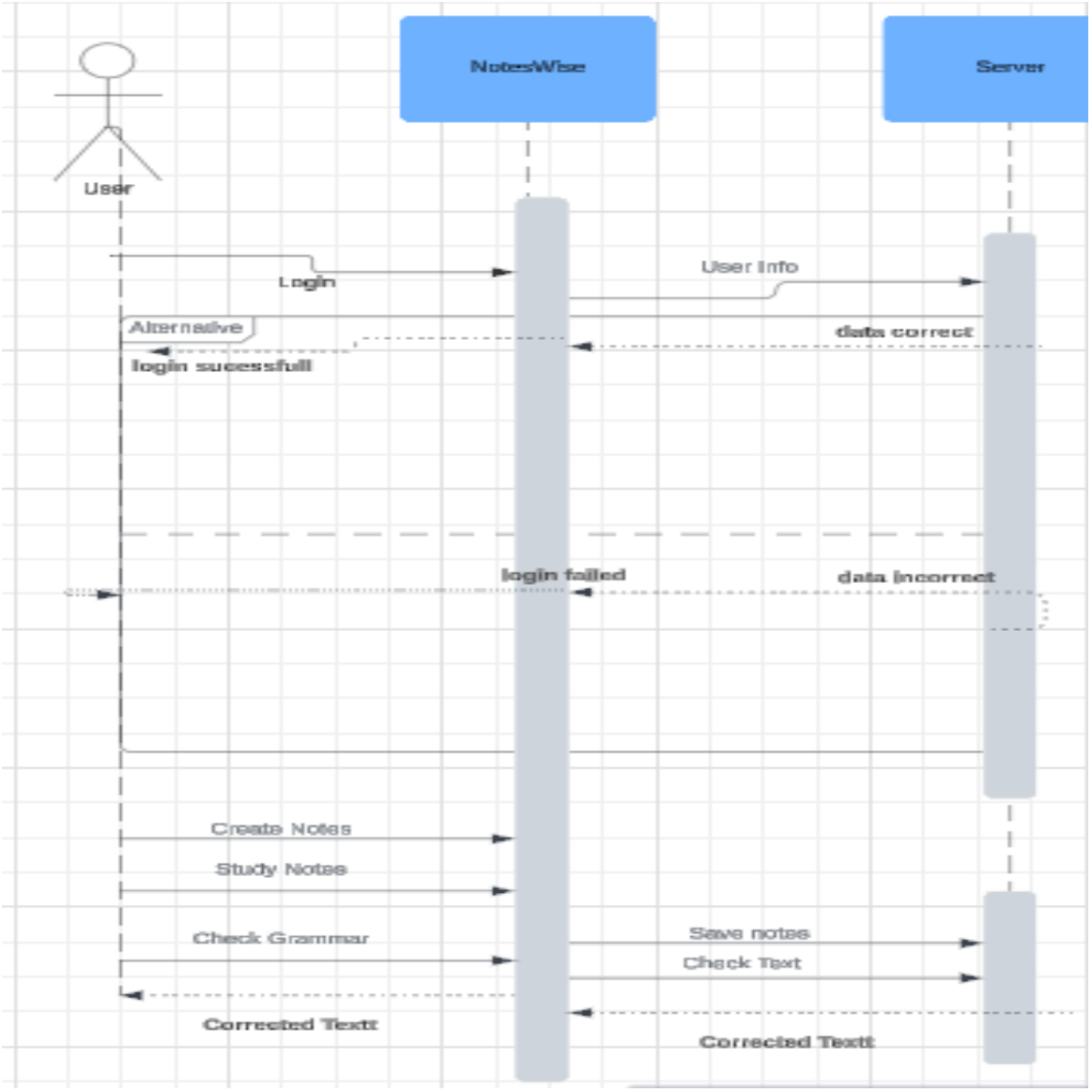


Figure 4: Sequence Diagram

0.7.5 System Design Diagram

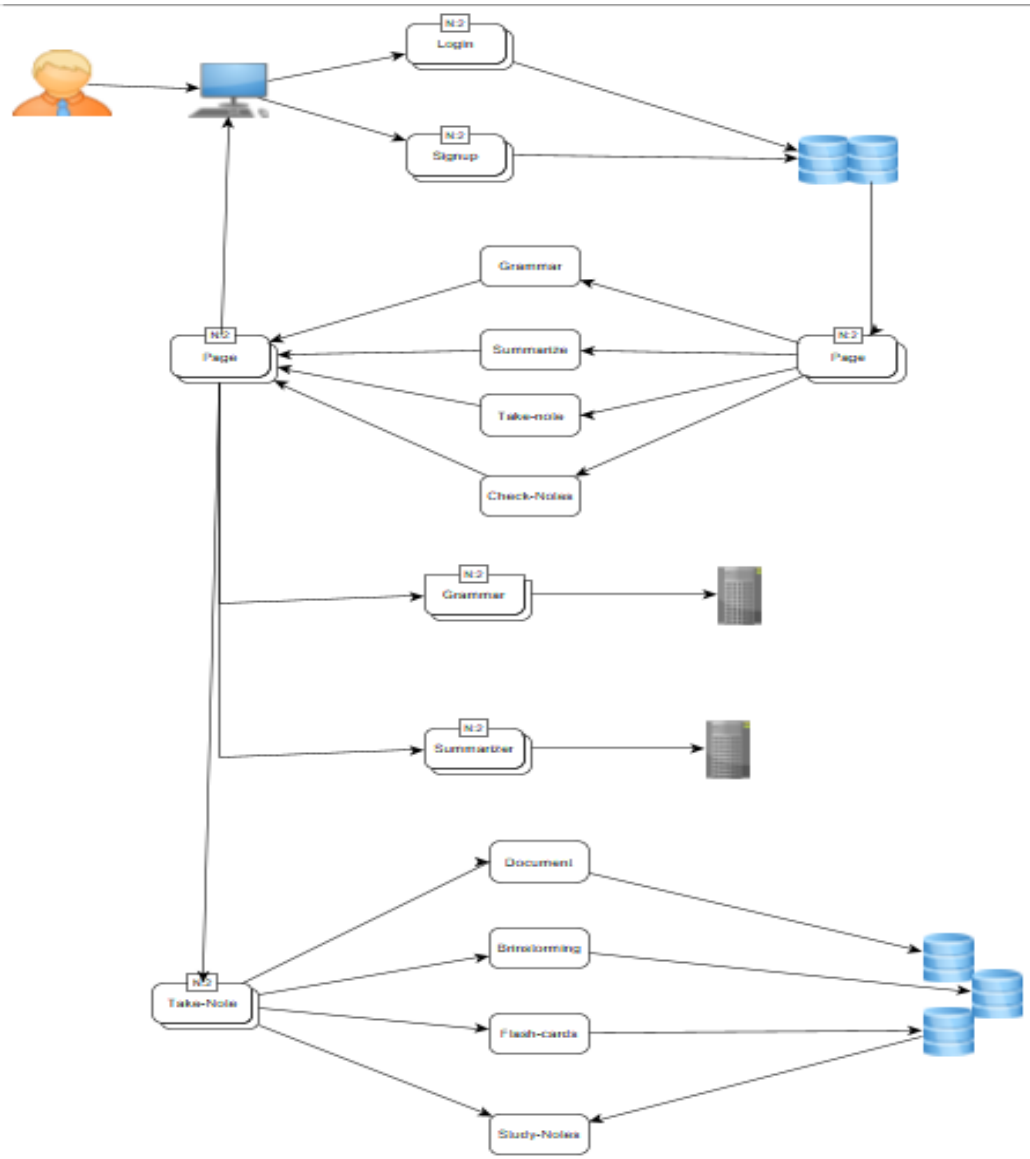


Figure 5: System Design Diagram

0.8 Project Timeline

0.8.1 Gantt Chart

Phase	Oct 2024	Nov 2024	Dec 2024	Jan 2025	Feb 2025	Mar 2025	Apr 2025	May 2025	Jun 2025	Jul 2025
Project Planning & Research	✔	✔								
System Design & Architecture		✔	✔							
Backend Development				✔	✔	✔				
Database Design & Setup					✔	✔				
Frontend Development				✔	✔	✔				
Testing & Quality Assurance						✔	✔			
Deployment & Documentation							✔	✔		
Final Review & Project Presentation								✔	✔	

Figure 6: Gantt-chart

Software Components

0.9 Development Framework and Tool

0.9.1 Tool

Vs Code:Visual Studio Code is a lightweight, open-source code editor developed by Microsoft, offering powerful features like syntax highlighting, debugging, Git integration, and support for a wide range of programming languages through extensions.

0.9.2 Framework

Django: Django is a high-level Python web framework that encourages rapid development and clean, pragmatic design, providing built-in tools for database handling, user authentication, and scalable web application development.

0.10 Python and Js Libraries

0.10.1 Js Libraries

A. **Konva.js**: Konva.js is a 2D canvas JavaScript library that enables the creation of complex, interactive graphics and animations, commonly used for applications like drawing editors, game UIs, and custom visualizations.

B. **Jodit.js**: Jodit.js is a lightweight, WYSIWYG rich text editor for JavaScript that offers a clean interface, inline editing, and a wide range of formatting tools for building user-friendly content editing experiences.

0.10.2 Python Libraries

A. **Google Translate API**: A cloud-based translation service that supports automatic translation between over 100 languages.

B. **Wikipedia API**: Used to fetch multilingual content and definitions from Wikipedia for NLP and knowledge-based tasks.

C. **WordNet**: A lexical database of English that groups words into sets of synonyms and provides definitions and relationships.

D. **TensorFlow**: An open-source machine learning framework by Google, widely used for training neural translation models.

E. **Pandas**: A Python library for data manipulation and analysis, useful for preprocessing text datasets in NLP projects.

F. **NumPy**: A library for numerical computing in Python, often used for handling vectors and matrices in NLP pipelines.

G. **NLTK (Natural Language Toolkit)**: A comprehensive Python library for text processing, including tokenization, parsing, and tagging.

H. **spaCy**: An industrial-strength NLP library with fast tokenization, POS tagging, named entity recognition, and dependency parsing.

I. **Transformers (by Hugging Face)**: A popular library for using pre-trained transformer models like BERT and T5 for translation and other NLP tasks.

J. **Sumy**: A simple library for extractive text summarization, often used in NLP tasks related to content reduction.

0.11 AI Models

0.11.1 Grammar Model

- **Model:** `MRNH/mbart-english-grammar-corrector` — A transformer-based grammar correction model fine-tuned on English text using Facebook's mBART architecture. It is capable of detecting and correcting common grammatical mistakes in user-generated content.

0.11.2 Summarization Models

- **Model 1:** `humarin/chatgpt_paraphraser_on_T5_base` — Based on the T5 model architecture, this model performs sentence-level paraphrasing and simplification, making text more readable while preserving its original meaning.
- **Model 2:** `vmkhoa2000/text_summarization-cnn` — A sequence-to-sequence model trained on the CNN/DailyMail dataset for generating concise summaries from longer texts, supporting both extractive and abstractive summarization.

0.11.3 QuizCraft Models

- **Model 1 (QA):** `distilbert-base-uncased-distilled-squad` — A smaller, faster version of BERT optimized for question answering, pre-trained and fine-

tuned on the SQuAD dataset to extract relevant answers from provided contexts.

- **Model 2 (QG):** `allenai/t5-small-squad2-question-generation` — A T5-based model trained on SQuAD2 for generating questions given a passage and answer, supporting automated quiz and assessment generation.

0.11.4 Chatbot Model

- **Model File:** `model.h5` — A custom-trained neural network stored in HDF5 format, developed for simple rule-based or intent-driven conversational responses in a chatbot interface.

Results and Discussion

0.12 Testing and Results

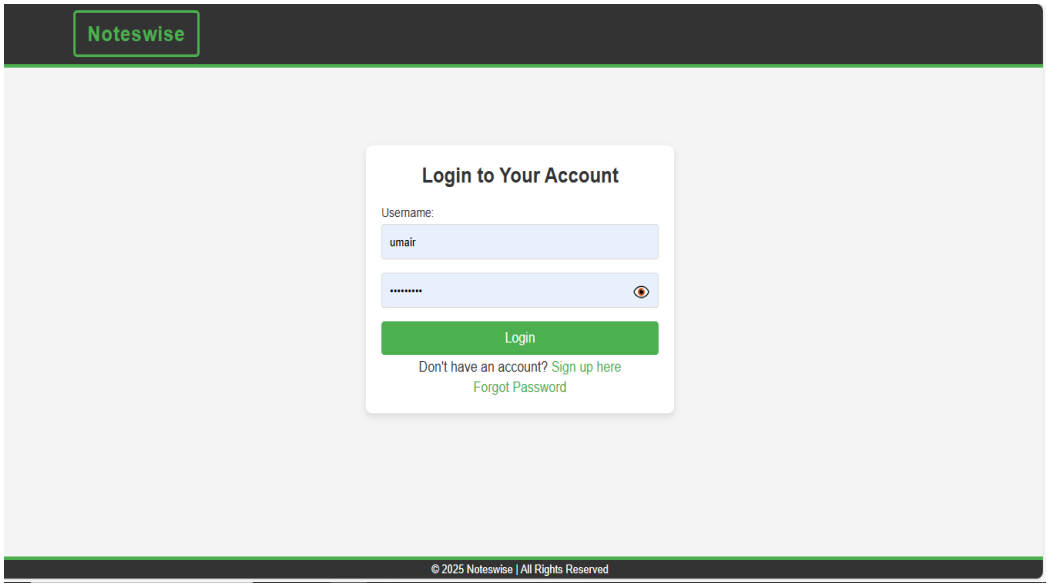


Figure 7: Authentication Page

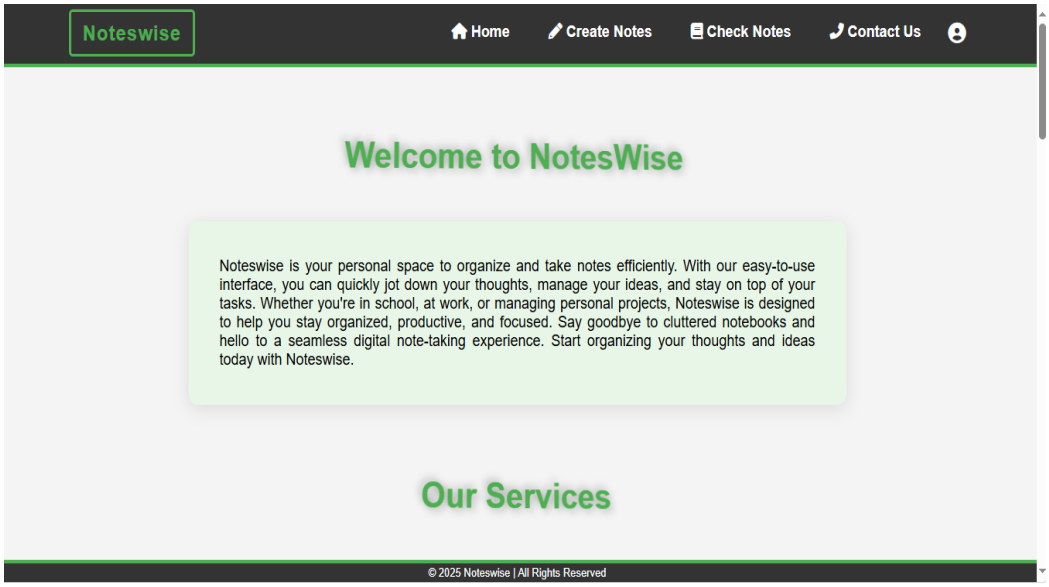


Figure 8: Home Page

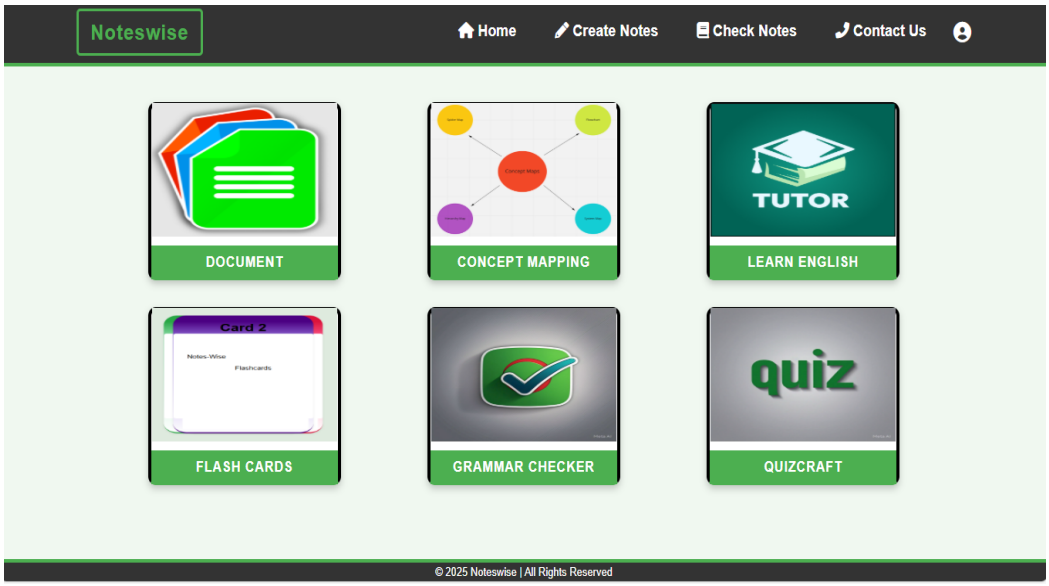


Figure 9: CreateNotes Page

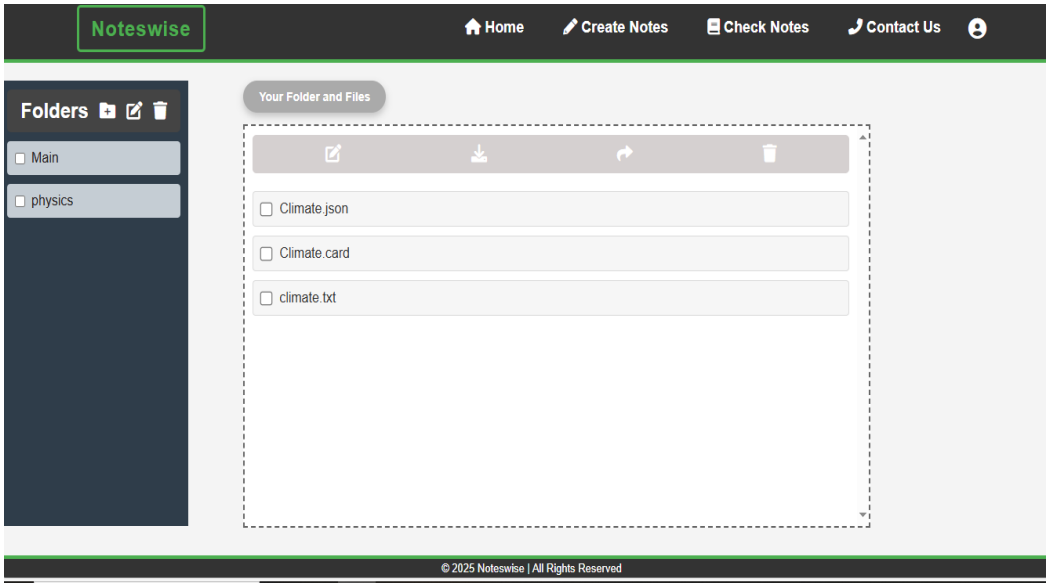


Figure 10: CheckNotes Page

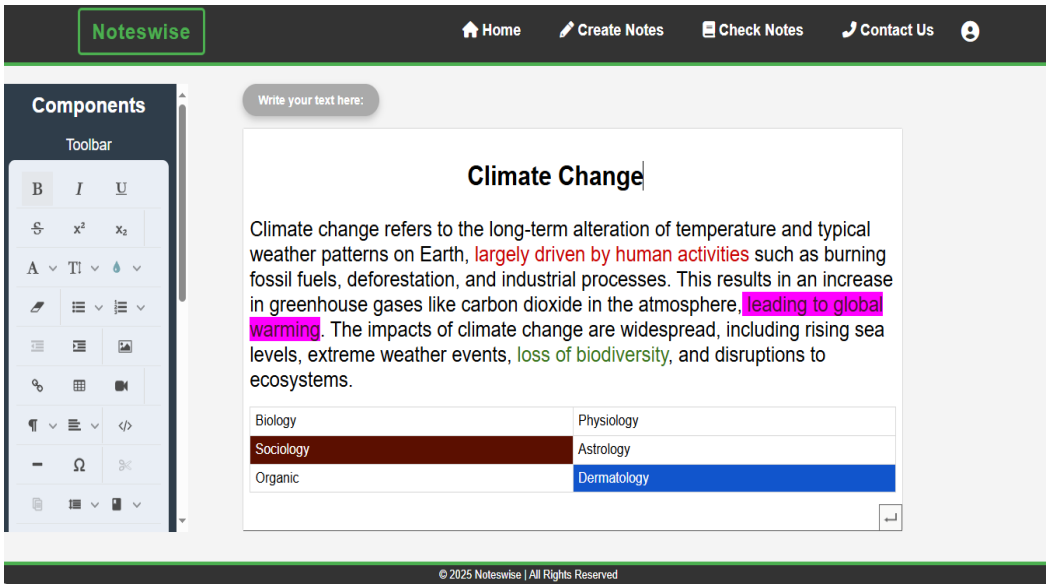


Figure 11: Document Page

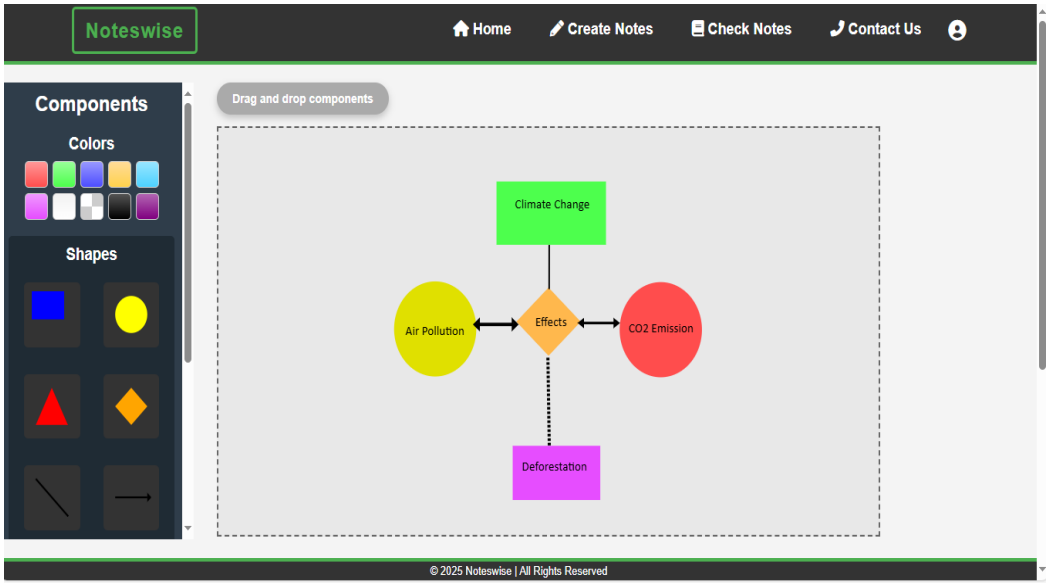


Figure 12: Diagram Page



Figure 13: FlashCards Page

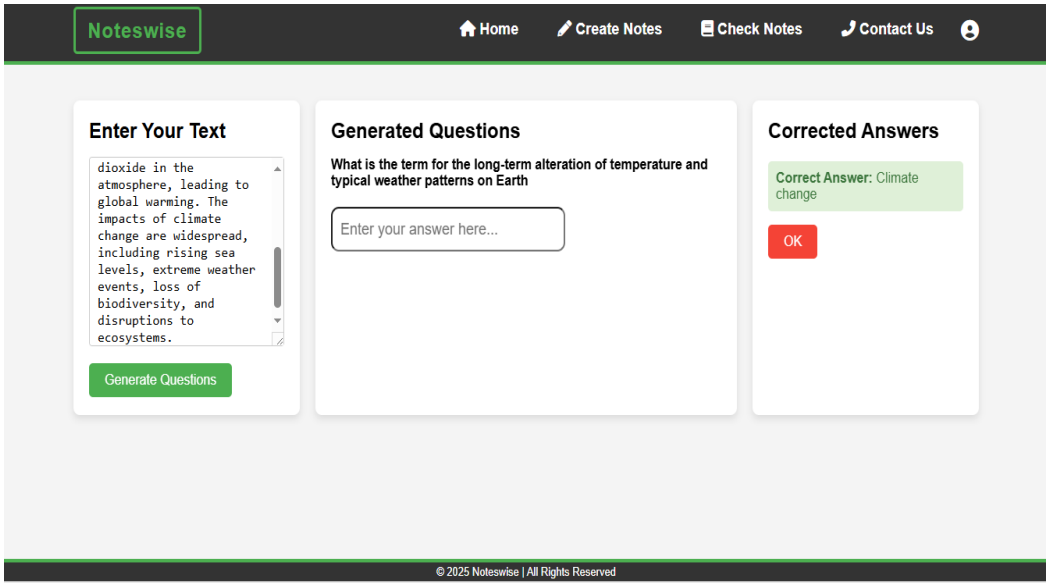


Figure 14: QuizCraft Page

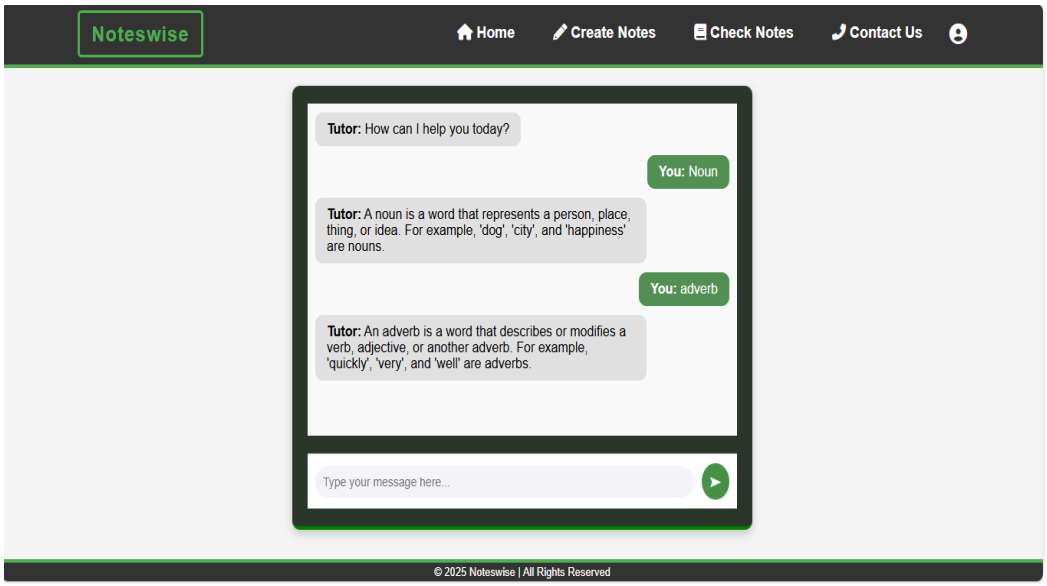


Figure 15: ChatBot Page

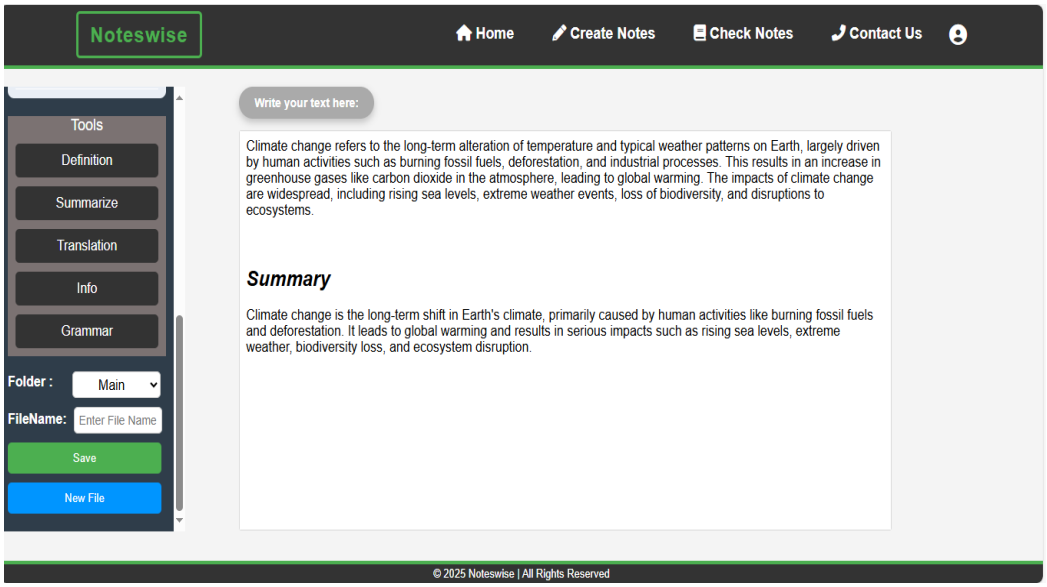


Figure 16: Summarizer Page

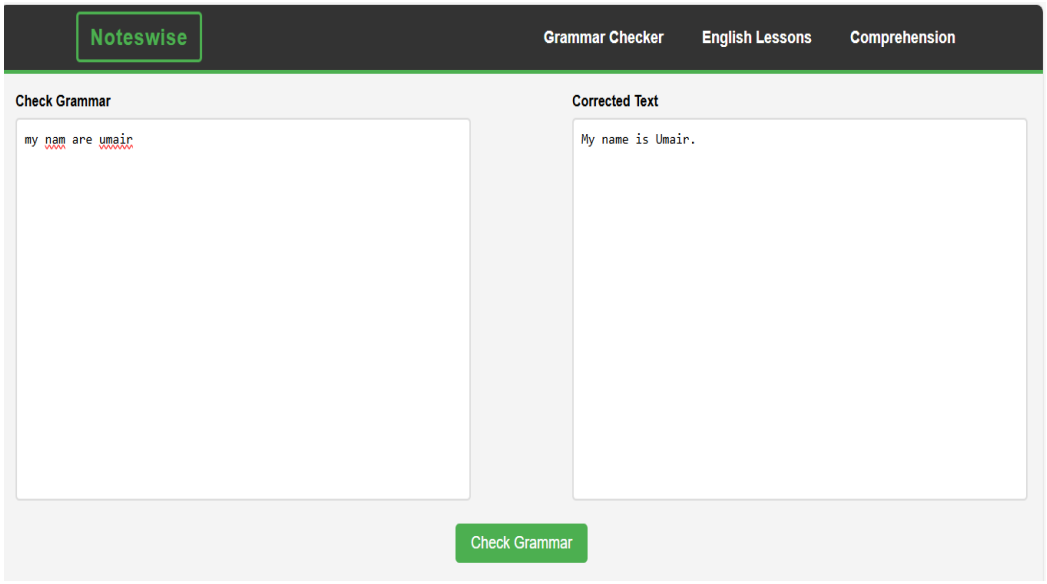


Figure 17: Grammar Result

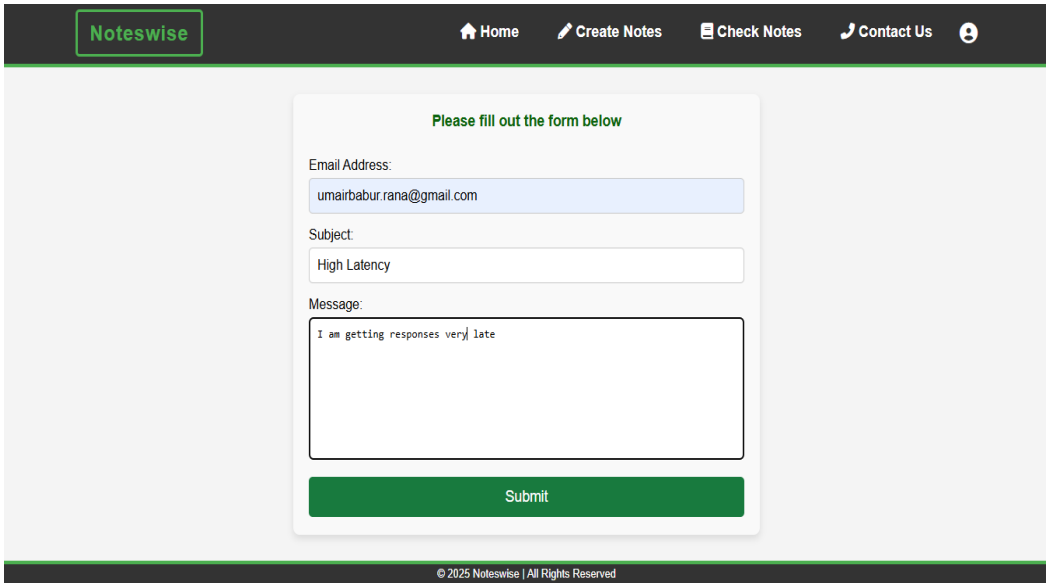


Figure 18: ContactUs Page

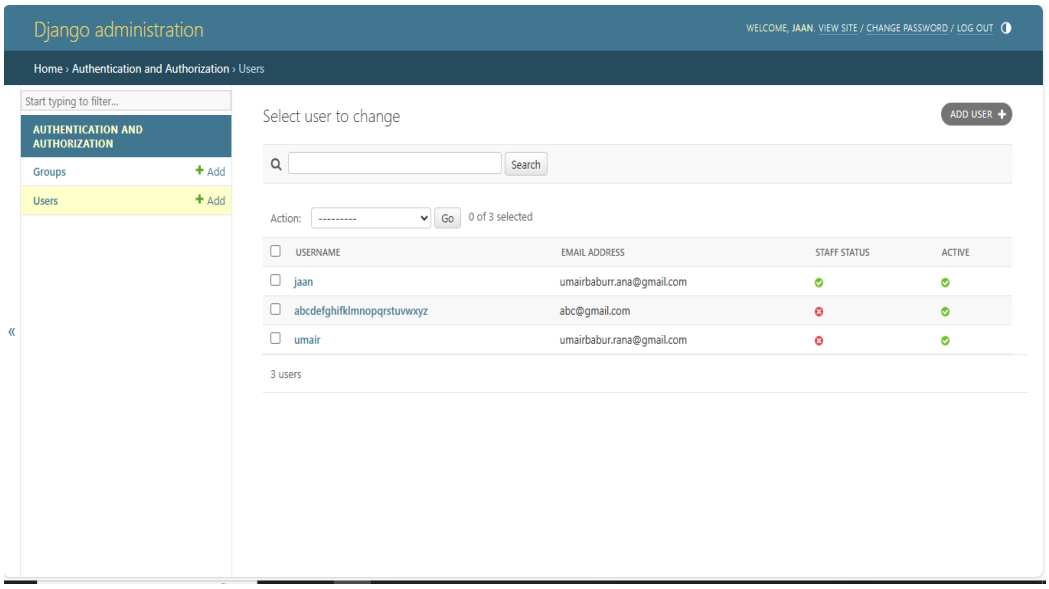


Figure 19: Admin Page

0.13 Discussion and Recommendations

0.13.1 Discussion

The development of the **NotesWise** system has successfully introduced a comprehensive, user-friendly platform for creating, managing, and interacting with multiple types of notes. The system supports textual, diagrammatic, and flashcard-based learning, along with intelligent tools like grammar checking, summarization, and quiz generation. Integration with Django and PostgreSQL provided a robust backend to support secure user data handling and real-time updates.

During implementation, significant attention was given to user experience, ensuring the interface remained intuitive across both desktop and mobile devices. Challenges such as mobile drag-and-drop behavior, shape alignment, and cross-device compatibility required tailored frontend solutions, particularly for the Konva.js canvas system.

The chatbot and AI tools significantly enhance the learning experience by providing interactive educational assistance. However, these AI-driven features also introduce dependencies on external APIs and require thoughtful error handling and scalability planning.

0.13.2 Recommendations

Based on current findings and future goals, the following recommendations are made to guide ongoing development.

- **Improve Accessibility:** Add support for screen readers, keyboard navigation, and voice commands to make the platform more inclusive.

- **Add Version Control:** Introduce a note versioning system that allows users to track changes, view revision history, and restore previous versions.
- **Enhance AI Reliability:** Optimize grammar, summarization, and quiz models for offline or limited-connectivity environments using local inference or lightweight models.
- **User Analytics:** Provide dashboards that give users insights into their study patterns, quiz performance, and note-taking trends for self-improvement.
- **Security Improvements:** Implement two-factor authentication (2FA), data encryption at rest, and regular security audits to protect user data more rigorously.
- **Feedback Loop:** Incorporate user feedback mechanisms within the app to allow continuous improvement based on real-world usage and suggestions.

These recommendations aim to not only strengthen the current platform but also prepare it for broader adoption in educational environments such as schools, colleges, and training institutes.

Use-Case Catalogue

This chapter enumerates every functional use-case of the **NotesWise** system. The use-cases are grouped under thematic headings for clarity. Each item is followed by a concise four-sentence description that defines its goal, scope, and primary flow. Together they form the foundation for requirements, design, and test planning.

0.14 Authentication

User Sign Up

A visitor registers by submitting a unique e-mail, secure password, and optional profile data. The system validates format and uniqueness, then persists the account and a default *main* workspace. A verification or welcome message is dispatched to confirm ownership. Successful completion redirects the new user to the login page.

User Login

A registered user supplies valid credentials on the login form. The system authenticates against hashed records and initiates a session on success. Session cookies maintain identity across requests while observing security best practices. The user is taken directly to their personal dashboard or the most-recent editor view.

User Logout

While authenticated, the user selects the logout option from any screen. The server invalidates the session token and purges cached user context. The browser's authentication cookie is deleted to prevent reuse. The user is returned to the public landing page with a confirmation flash message.

Password Reset

A user who cannot recall their password requests a reset link. The system generates a time-limited token and e-mails it to the registered address. Hitting the link presents a secure form to set a new password that meets strength rules. The account is updated, the token is revoked, and the user may now log in normally.

0.15 Workspace & File Management

Create Folder

Within their workspace the user invokes *New Folder*. They supply a unique name; the system stores a folder record linked to the user. The folder appears in the sidebar hierarchy in alphabetical order. The user can now save or move notes into it.

Rename Folder

The user selects a folder and chooses *Rename*. A modal prompts for a new title, checked for uniqueness within the same depth. On confirmation the folder record is updated and the UI refreshes instantly. No child-note linkage is affected beyond path name changes.

Delete Folder

The user chooses *Delete* on a non-empty folder. The system asks for confirmation and warns that contained notes will also be removed or moved. If confirmed, it cascades deletes (or relocates to *main*) and updates storage quotas. A success toast notifies completion.

View All Folders

Clicking *Folders* loads a list or tree of every folder owned by the user. The UI shows counts of contained notes and last-modified timestamps. Server pagination ensures performance for large sets. Selecting any folder filters the main note pane accordingly.

Create Note

Inside a folder, the user presses *New Note* and chooses text, diagram, or flashcard. The appropriate editor opens in a blank state with default metadata. A draft is autosaved locally until the first explicit save to the database. The note record stores owner, folder, type, and timestamps.

Edit Note

From any listing the user clicks a note title to open its editor. Current content is fetched from the database and rendered for interaction. Changes are persisted on save or periodically via autosave. Versioning may record deltas for future rollback.

Delete Note

In the editor or list view the user triggers *Delete*. A confirmation dialog guards against accidental loss. Upon approval the note row is soft-deleted (or hard-deleted if policy allows). The list view refreshes and storage metrics update.

Move Note Between Folders

Drag-and-drop (or context menu) lets a user move a note to another folder. The client sends the new folder ID; the server updates foreign-key references. Folder counts, breadcrumbs, and URLs update instantly via websockets. No content is altered beyond its path.

View Notes by Folder

Selecting a folder filters the central table to notes in that folder only. Sorting, search, and pagination apply within the active subset. The URL reflects the folder for shareable deep links. Switching folders re-queries to keep data current.

View All Notes

Choosing the *All Notes* filter loads every note the user owns. A combined list appears with icons for type and folder location. Users may apply global search across titles and content snippets. This view aids housekeeping and bulk actions.

0.16 Text Notes

Create Text Note

The user opens the rich-text editor to compose prose, equations, or code. Markdown shortcuts and toolbar controls format the content. Metadata such as tags and title are set in a sidebar panel. First save commits the note to the server and returns an ID.

Edit Text Note

Revisiting a text note loads its stored HTML or Markdown into the editor. Users modify content, apply styles, or embed media via drag-and-drop. Autosave protects against data loss every few seconds. Closing the editor triggers a final flush and clears the draft cache.

Save Text Note

Saving validates that content is not empty and that title uniqueness is respected within the folder. The server stores the updated body, metadata, and a revision number. A success snackbar affirms completion and updates the timestamp. Subsequent edits increment the revision for audit trails.

View Text Note

Selecting a text note in read-only mode renders content with proper styling. The interface hides editing controls for distraction-free reading. A table of contents provides quick navigation for long documents. A button toggles into full edit mode if the user

has write access.

Delete Text Note

The deletion flow mirrors the generic note delete but also removes any embedded assets. Attachments referenced solely by the note are queued for garbage collection. The user is informed how many linked resources will be affected. A backup may remain in the trash bin until purged by policy.

0.17 Diagram Notes

Create Diagram

The user opens a canvas powered by Konva.js with drag-and-drop shape palette. They place circles, rectangles, and connectors to form concept or mind maps. Shape positions and attributes are serialized to JSON on every change. Saving stores the JSON and a preview thumbnail to the database.

Save Diagram (JSON)

When the user hits *Save*, the canvas exports its layer tree. The JSON captures node types, coordinates, styles, and z-order. The server records this alongside metadata and a PNG thumbnail. Revisioning enables future diff and restore capabilities.

View/Edit Diagram

Opening a diagram pulls the JSON and rehydrates it into the canvas. Users can move, resize, or delete shapes using touch or mouse gestures. The transformer tool shows

bounding handles for precise manipulation. Changes commit exactly as in creation, preserving history.

Delete Diagram

Deletion follows the standard note pattern but additionally wipes associated thumbnails. The JSON blob is removed and storage is reclaimed. Shared links become invalid post-deletion. Activity logs record the action for auditing.

0.18 Flashcards

Create Flashcard Set

The user chooses *New Flashcards* and gives the set a title. A grid appears where front and back text are entered for each card. The set saves as a single note with JSON array entries. A progress bar tracks completion percentage.

Add/Edit/Delete Cards

Within an existing set the user can append new cards, edit text, or remove entries. Inline validation ensures no card is blank before save. The JSON array is patched and versioned per update. The UI re-indexes card positions after every change.

View Flashcard Set

Selecting a set opens the study mode with one card in focus. Arrow keys or swipe gestures flip between cards, hiding the opposite side. A sidebar shows progress and allows shuffling order. Completion stats update in real time.

Practice Flashcards

In practice mode each card starts front-side up; the user flips to see the answer. Spaced-repetition scheduling can mark cards as *Easy*, *Hard*, or *Again*. The system records outcomes to compute the next review date. A summary screen shows mastery levels at session end.

0.19 Smart Note Tools

Grammar Check

A user triggers grammar analysis while editing a text note. The client sends plain text to an NLP micro-service for checking. Detected issues are highlighted inline with suggestions. Accepting a suggestion rewrites the affected span.

Summarize Text

Clicking *Summarize* sends the note to an abstractive summarizer. A condensed paragraph or bullet list returns and is shown in a sidebar. Users may copy it into a new note or replace content. Summaries respect user privacy by staying within the tenant boundary.

Quiz Generation

The user chooses *Generate Quiz* for study reinforcement. The system extracts key points and forms MCQs or short-answer questions. A quiz note is created, linked back to the source for context. Users can edit questions before saving.

0.20 AI Chatbot

Study Questions

A learner poses concept questions in natural language. The chatbot retrieves or synthesizes concise explanations with references. Follow-up prompts allow deeper exploration or examples. The dialogue context remains for multi-turn learning.

Definitions & Explanations

Highlighting a term in a note and pressing *Ask AI* queries the bot. A plain-language definition plus cross-references returns in a tooltip. Citations point to reputable sources or internal materials. The user may insert the answer directly into their note.

0.21 Cloud Storage & Access

Save Workspace

Each save operation persists data to cloud storage with redundancy. Atomic transactions guarantee notes never partially save. Compression and deduplication optimize space usage. Backups run nightly with point-in-time recovery.

Access Any Device

The web app is responsive and offers PWA offline caching. Authenticated users can open their workspace from mobile, tablet, or desktop. Data syncs automatically once connectivity is restored. Device fingerprints help secure unfamiliar logins.

Sync Across Devices

Real-time websockets reflect edits made on another device instantly. Conflict-resolution merges concurrent changes or alerts the user. Batch sync jobs reconcile background edits after offline periods. A status indicator shows last sync success.

Future Tasks

This chapter outlines planned enhancements and feature expansions for the **NotesWise** system. These tasks aim to enrich user experience, extend accessibility, and incorporate more intelligent study tools. Each item is currently under consideration or prototyping and will be prioritized based on user feedback and technical feasibility. Future updates will integrate these seamlessly into the existing architecture.

0.22 Audio Notes

0.22.1 Record and Save Audio

Users will be able to record audio directly within the note interface using a built-in microphone tool. The audio file will be saved securely in the user's storage with appropriate metadata such as duration and timestamp. Each audio note may optionally include a title and description for quick reference. Playback controls will be available in the editor and viewer interfaces.

0.22.2 Convert Audio to Text

Users may convert recorded or uploaded audio into transcribed text using speech recognition services. The transcript will appear alongside the audio file or as a separate

note. Users can edit the transcription to correct any misinterpretations. This feature will enhance accessibility and support voice-based note-taking.

0.23 Collaborative Notes

0.23.1 Real-Time Collaboration

Multiple users will be able to edit a shared note simultaneously. Changes will sync in real time with color-coded cursors and edit markers. Permissions such as read-only or comment-only access will be configurable. This will support group study, co-authoring, and peer review.

0.24 Note Tagging & Search Filters

0.24.1 Tag Notes

Users will be able to assign custom tags to any note for easier classification. Tags will appear as chips or labels in the UI and can be added during editing or from the list view. Suggested tags may be generated based on content analysis. A sidebar will list all tags for quick filtering.

0.24.2 Advanced Search

A redesigned search interface will support queries by title, content, tags, date, and note type. Filters and sorting options will enhance discoverability of older or forgotten notes. Search results will highlight matches with preview snippets. This will greatly improve navigation in large workspaces.

0.25 Daily Planner Integration

0.25.1 Create Study Schedule

Users will be able to create a daily or weekly planner within the app. Tasks like reviewing flashcards, editing notes, or practicing with the chatbot can be scheduled. Reminders and notifications will keep users on track. The planner view will integrate with notes to link tasks directly to resources.

0.26 Offline Editing Mode

0.26.1 Local Editing with Sync

Users will be able to edit notes while offline using a progressive web app (PWA) framework. Changes will be cached locally and synchronized when connectivity is restored. Conflict handling will manage simultaneous edits from multiple devices. A sync status icon will inform users of pending updates.

Conclusion and Bibliography

0.27 Conclusion

The development of the Notes-Wise web application significantly improves the note-taking experience and enhances English writing skills [11]. Notes-Wise addresses the limitations of traditional note-taking systems. The application enables users to create, manage, and store notes effectively, while providing real-time grammar correction and educational resources.

The combination of functional and non-functional requirements, along with a user-centered design approach, ensures that the application meets the needs of a diverse audience, including students, professionals, and language learners. By leveraging modern web technologies, we delivered a scalable and reliable product.

Notes-Wise aims to support users in their educational and professional endeavors, fostering a more organized and productive learning environment. Future enhancements, including additional language support and mobile accessibility, will further strengthen the app's role in improving writing skills and productivity.

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